

INVESTMENT OPPORTUNITY REPORT

Nigeria Power and Energy Sector

Financing the Root Crisis: A Case for Private Capital in Nigeria's Energy Infrastructure

Prepared by
Tolu Adetuyi, MBA
Co-Founder and Chief Innovation Officer
Prembly Inc.

Report Date
March 2026
Classification
Confidential

\$100bn+ Power Sector Deficit	4GW / 30GW+ Delivered vs. Required	80 million+ Nigerians Without Electricity	15% to 25%+ Target Equity IRR
---	--	---	---

SECTION 01

Executive Summary

Nigeria is the fourth largest economy in Africa, with a nominal GDP of approximately \$300 billion and a population exceeding 200 million. Yet the country's productive capacity is fundamentally constrained by a single, persistent failure: the inability to generate, transmit, and distribute adequate electricity to its citizens, businesses, and institutions.

The power sector deficit sits at over \$100 billion in unmet infrastructure need. The national grid delivers fewer than 4 gigawatts of electricity on most days against a suppressed demand of at least 30 gigawatts. Over 80 million Nigerians have no access to electricity at all. Those who do face an unreliable supply that averages fewer than 12 hours per day in most commercial areas. In response, the private sector collectively spends an estimated \$12 to \$14 billion annually on diesel generators to maintain basic operations. That expenditure produces no productive asset, generates no return, and compounds year over year.

This report makes a straightforward case: the power crisis is not primarily a technical problem. It is a capital allocation and deal-structuring problem. The technology exists. The policy frameworks are improving, anchored by the landmark Electricity Act of 2023. The demand is unambiguous and backed by demonstrated willingness to pay. What is absent is the structured deployment of private capital at scale.

This report identifies five investment models within the power sector that are viable today, executable at multiple entry scales, and capable of delivering equity returns between 15 and 35 percent depending on the model and structure. It also briefly identifies four adjacent sectors where energy access is the primary constraint, and whose productivity gains will compound the returns available to power investors.

If Nigeria's electricity problem is solved, it does not merely improve one sector. It unlocks the productive capacity of every other sector simultaneously. Power is the master variable in Nigeria's economic equation.

SECTION 02

The Power Sector: A Structural Diagnosis

2.1 The Generation Gap

Nigeria's installed electricity generation capacity stands at approximately 13 gigawatts across thermal and hydropower plants. In practice, the grid delivers between 3,500 and 5,000 megawatts on any given day. The difference between installed and delivered capacity is accounted for by gas supply disruptions, aging infrastructure, transmission constraints, and persistent commercial losses across the value chain.

The government's target is 30 gigawatts of accessible generation capacity by 2030. Reaching that figure requires an investment of over \$100 billion across generation, transmission, and distribution infrastructure. Current budgetary allocations, which amount to approximately 5 percent of the stated annual requirement, make public funding alone categorically insufficient. Private capital is not an option in this equation. It is a necessity.

Indicator	Current Status	Implication for Investors
Installed generation capacity	13 GW	<i>Significant headroom exists between installed and delivered capacity</i>
Electricity actually delivered	3.5 to 4.0 GW	<i>Structural delivery failure creates demand for off-grid and embedded generation</i>
Population without electricity	80 million+	<i>Captive, underserved market with demonstrated willingness to pay</i>
Annual private diesel spend	\$12 to \$14 billion	<i>Confirms price tolerance; any solution cheaper than diesel is immediately competitive</i>
Government budget allocation	5% of annual need	<i>Public funding gap is structural; private capital is the only viable bridge</i>
Revenue collection improvement	40% rise in 2024	<i>Tariff reform is working; collection environment is materially improving</i>

2.2 The Policy Turning Point: Electricity Act 2023

The Electricity Act of 2023 is the most consequential regulatory reform in Nigeria's power sector in two decades. It decentralised authority over electricity markets, empowering state governments to license, regulate, and develop independent power systems without federal approval. This single structural change removes the primary bureaucratic bottleneck that stalled private sector participation for years.

State governors can now serve as direct off-takers for power purchase agreements. Lagos, Rivers, Kano, Ogun, and Oyo, all of which have significant commercial and industrial activity, can anchor

solar and embedded generation projects under commercially negotiated terms. This creates a new class of creditworthy counterparty for private developers and investors.

Complementing the Electricity Act, the National Integrated Electricity Policy endorsed in 2025 aligns federal and state priorities, while tariff adjustments in 2024 raised Band A electricity tariffs to N225 per kilowatt-hour, improving revenue collection by 40 percent within the year. These are not cosmetic reforms. They represent a structurally improved investment environment.

The Electricity Act 2023 changed the answer to the question: 'Who is the off-taker?' Previously, that answer was a single, struggling federal utility. Today, it can be a state government, a university, a hospital, a manufacturer, or a commercial estate. The risk profile of the sector has fundamentally changed.

2.3 The Diesel Economy as Market Proof

Nigeria's private sector spends between \$12 and \$14 billion annually on diesel-powered generators. At an average cost of \$0.35 per kilowatt-hour, this is the most expensive form of electricity consumption in any large economy in the world. It is also the clearest possible evidence that demand exists and that the market will pay.

Any investor offering electricity at \$0.10 to \$0.18 per kilowatt-hour, which is the realistic price range for solar-generated power under a commercial power purchase agreement, is immediately competitive with diesel. The customer does not need to be convinced to switch. The switching decision is self-evident once the alternative is structured and available. This is a rare investment context: the demand is proven, the willingness to pay is established, and the competing product is more expensive, noisier, more polluting, and less reliable than what a well-structured solar solution can offer.

SECTION 03

Investment Models: Five Viable Entry Points

The power sector is not a monolithic investment proposition. It disaggregates into five distinct models, each with a different capital requirement, risk profile, revenue structure, and return expectation. An investor entering this space does not need to build a power plant. There is a viable

entry at every scale, from \$200,000 to \$200 million, and each model has documented precedents in the Nigerian market.

Model 1 | Solar Mini-Grids

Entry Scale: \$500,000 to \$5 million per site

A solar mini-grid serves a defined cluster of residential, commercial, and institutional customers in areas where the national grid is absent or unreliable. The developer installs solar panels, battery storage, and a local distribution network, then sells metered electricity directly to end-users at regulated tariff rates. Revenue is collected via prepaid meters and payment platforms, eliminating credit risk at the customer level.

The Rural Electrification Agency provides licensing and, in certain cases, viability gap funding to mini-grid developers to improve project economics in lower-income areas. The government's target of achieving 30,000 mini-grid projects nationwide by 2030 creates a well-defined pipeline. Nigeria currently has fewer than 200 functional commercial mini-grids. The gap between 200 and 30,000 is not a policy ambiguity. It is an investment opportunity.

- **Revenue:** Revenue mode is metered electricity sales to residential and commercial customers at N50 to N120 per kilowatt-hour
- **Scale:** Average generation capacity per site: 50 to 500 kilowatts depending on community size and demand density
- **Return:** Project IRR is estimated around 18 to 25 percent depending on community size, REA viability gap funding, and operational cost management
- **Payback:** Payback period of 4 to 6 years
- **Primary Risk:** Key risk is customer payment discipline, managed through prepaid metering and payment integration

The proof of concept for this model exists. Power Africa's programme has facilitated over 1,005 megawatts of capacity and unlocked \$1.48 billion in investments for on-grid and off-grid projects in Nigeria. Multiple developers, including Husk Power Systems, Nayo Tropical Technology, and Rural Spark, have demonstrated viable unit economics in Nigerian mini-grid deployments. This is not an experimental model. It is a replicable one.

Model 2 | Commercial and Industrial Solar (Power Purchase Agreements)

Entry Scale: \$2 million to \$50 million

The Commercial and Industrial solar model addresses the single largest direct cost of doing business in Nigeria. A developer installs rooftop or ground-mounted solar generation at a factory, hospital, university, retail centre, or industrial estate, then sells the electricity generated to that facility under a long-term Power Purchase Agreement at a rate below what the facility currently pays for diesel-generated power. The customer saves money immediately. The developer earns a contracted cash flow for 10 to 25 years.

This model entirely bypasses the national grid. It does not depend on distribution companies, does not require wheeling agreements, and does not expose the investor to grid instability. The risk profile is anchored to the creditworthiness of the corporate off-taker, not the government or the utility sector. For a large bank, a cement manufacturer, a supermarket chain, or a pharmaceutical company, the credit risk is materially different from sovereign risk.

- **Pricing:** Electricity price to customer between \$0.10 to \$0.16 per kilowatt-hour, compared to \$0.30 to \$0.40 per kilowatt-hour for diesel
- **Contract:** Contract structure of 10 to 25 year Power Purchase Agreement with creditworthy corporate counterparty
- **Return:** Project IR of 18 to 24 percent; leveraged returns can exceed 30 percent with favourable debt structuring
- **Market:** Addressable market is every manufacturer, bank, hospital, hotel, and educational institution currently running a diesel generator, estimated at over 60 million generators nationally
- **Payback:** Payback period of 4 to 7 years

The Commercial and Industrial solar model is the cleanest risk proposition in Nigeria's energy sector today. The customer is a business that already pays for electricity. You are simply offering it to them at a lower price from a more reliable source. No subsidy required. No government dependency. No currency mismatch if contracts are structured in dollar-equivalent terms.

Model 3 | PAYGO Home Solar Systems

Entry Scale: \$1 million to \$20 million (portfolio financing)

Pay-as-you-go solar financing solves the access problem at the household level. A company provides a solar home system comprising panels, battery storage, and basic appliances to a household in exchange for small, regular payments made via payment apps. The system is remotely monitored and can be deactivated if payments lapse, eliminating the moral hazard of

non-payment. Once the customer completes payment, typically over 12 to 36 months, they own the system outright.

For investors, the proposition is a receivables portfolio. Investing in a PAYGO solar company means financing the receivables generated by household payment plans across thousands or hundreds of thousands of customers. The aggregated receivables are predictable, diversified, and secured against an asset that can be deactivated and recovered. The model has been successfully implemented at scale in Kenya, Tanzania, and Rwanda by companies such as M-KOPA and d.light, and is expanding in Nigeria.

- **Revenue:** Revenue mode is monthly repayments from household customers via payment platform, typically N3,000 to N8,000 per month
- **Return:** Target IRR for receivables investors is around 20 to 28 percent
- **DFI Access:** Development Finance Institution appetite such as DFC, IFC, and Norfund have all financed PAYGO portfolios in sub-Saharan Africa
- **Blending:** Blended finance opportunity for First-loss tranche from DFIs reduces effective risk and improves returns for commercial investors

Model 4 | Metering Infrastructure (Distribution Company Partnerships)

Entry Scale: \$1 million to \$30 million

Nigeria's 11 electricity distribution companies collectively under-meter their customer base by an estimated 40 to 60 percent. Unmetered customers are billed on estimated consumption, which suppresses collection, enables energy theft, and prevents accurate load management. The Meter Asset Provider regulatory framework, introduced by the Nigerian Electricity Regulatory Commission, enables private companies to finance, install, and own electricity meters that are then leased to distribution companies under shared-savings agreements.

A Meter Asset Provider installs prepaid smart meters at customer premises. The distribution company collects metered revenue and remits a monthly lease payment to the Meter Asset Provider. The federal government has committed to installing 1.1 million meters in the near term under the National Mass Metering Programme. The total metering gap is estimated at over 7 million connections. The gap between government programme targets and total need is the investment pipeline.

- **Revenue:** Revenue model is a monthly lease payments from distribution companies per meter installed

- **Return:** Target IRR is around 16 to 22 percent in naira terms, with hedging mechanisms available through indexed contracts
- **Policy:** Government supports through National Mass Metering Programme provides partial subsidy for low-income meter deployment
- **Payback:** Payback period is around 3 to 5 years

Model 5 | Utility-Scale Independent Power Projects

Entry Scale: \$50 million to \$500 million+

Utility-scale Independent Power Projects are the conventional infrastructure investment model in which a private developer constructs and operates a large generation facility, selling output to a utility, state government, or large industrial off-taker under a long-term Power Purchase Agreement. Nigeria has commissioned several utility-scale projects, including the Azura-Edo IPP and the Kainji hydropower privatisation. The Jigawa Solar PV initiative, currently in development, targets 50 to 100 megawatts in its initial phase with long-term expansion to 1,000 megawatts.

Utility-scale projects carry higher capital requirements and longer development timelines, but also attract the deepest pool of institutional capital, including pension funds, sovereign wealth funds, and major DFIs. The African Development Bank, IFC, and the US International Development Finance Corporation have all provided debt and equity financing to Nigerian IPPs. Blended finance structures, in which DFIs absorb first-loss risk, materially improve the economics for commercial equity investors.

- **Return:** Target equity IRR is around 14 to 18 percent for well-structured projects with creditworthy off-takers
- **Structure:** Debt financing, typically 60 to 70 percent leverage from DFIs and commercial banks
- **Timeline:** Development timeline estimated around 24 to 48 months from financial close to commercial operation
- **Risk:** Key requirement is a Bankable Power Purchase Agreement with sovereign guarantee or creditworthy state government as off-taker

SECTION 04

Consolidated Return Summary

The table below summarises the five investment models described in this report across their key financial parameters. Returns are expressed as equity Internal Rate of Return under well-structured commercial or blended finance arrangements. These figures reflect documented market data from comparable projects across Nigeria and sub-Saharan Africa.

Investment Model	Entry Capital	Equity IRR	Payback Period
PAYGO home solar (portfolio)	\$1M to \$20M	20% to 28%	3 to 5 years
Commercial and Industrial solar (PPA)	\$2M to \$50M	18% to 24%	4 to 7 years
Solar mini-grid (per site)	\$500K to \$5M	18% to 25%	4 to 6 years
Metering infrastructure (MAP)	\$1M to \$30M	16% to 22%	3 to 5 years
Utility-scale IPP (50MW+)	\$50M to \$500M+	14% to 18%	7 to 10 years

Returns at the lower end of each range reflect conservative assumptions: higher capital costs, moderate utilisation, and standard commercial financing. Returns at the upper end reflect well-structured blended finance arrangements with DFI first-loss support, optimised capital stacks, and experienced operational teams.

SECTION 05

Adjacent Sectors: Unlocked by Energy Access

The power sector is the enabling variable for Nigeria's broader infrastructure recovery. The table below identifies four sectors where electricity is the primary binding constraint. Resolving the energy deficit does not merely improve these sectors incrementally. In several cases, it transforms their viability entirely.

Sector	Financing Gap	How Energy Access Unlocks It
Affordable Housing	\$46.7 billion	Reliable power makes high-density urban housing viable, reduces cement and construction costs, and enables

Sector	Financing Gap	How Energy Access Unlocks It
		property technology platforms that depend on consistent connectivity.
Cold Chain and Agri-Logistics	\$4.5 billion	Cold storage is entirely electricity-dependent. Nigeria loses 40 percent of perishable food production annually due to absence of cold chain infrastructure. Affordable, reliable power makes refrigerated storage economically viable for the first time in secondary cities.
Digital Infrastructure	\$15 billion	Data centres, fibre repeater stations, and mobile towers all require stable power. Nigeria's data centre sector cannot scale to its projected 400 MW requirement without a credible electricity supply. Energy and digital infrastructure investment are inseparable.
Transport and Logistics	\$40 billion	Electric freight vehicles, automated port and warehouse systems, and rail electrification all depend on grid stability. Logistics cost reduction in Nigeria is materially linked to the cost and reliability of power for cold storage, warehousing, and vehicle charging.

Each of the sectors above represents a standalone investment thesis with compelling return profiles. They are noted here not to dilute the focus of this report, but to establish the compounding logic of energy investment: a dollar deployed in Nigerian power infrastructure does not earn a single-sector return. It earns a cross-sector multiplier.

SECTION 06

Risk Considerations and Mitigants

No investment report on Nigerian infrastructure is complete without an honest account of the risks. The following risks are material, well-understood by experienced market participants, and manageable through appropriate deal structuring.

- **Currency Risk:** Currency risk is the primary concern for US dollar investors. Naira depreciation reduces the dollar value of naira-denominated revenues.
- **Mitigant:** Structure Commercial and Industrial PPAs in dollar-equivalent terms indexed to the official exchange rate, or pair investments with dollar-revenue generating assets such as export-facing businesses.

- **Counterparty Risk:** Counterparty risk exists where government entities or distribution companies are the off-taker.
- Mitigant: Prioritise creditworthy private sector counterparties at entry stage; access World Bank partial risk guarantees for projects with government off-takers.
- **Regulatory Risk:** Regulatory and policy risk remains despite improvements. The regulatory environment, while materially better than five years ago, can shift with changes in administration.
- Mitigant: Engage international arbitration clauses in all project agreements; partner with DFIs whose participation serves as a political risk deterrent.
- **Execution Risk:** Operational execution risk in Nigeria's environment requires experienced local partners.
- Mitigant: Do not attempt to operate remotely. Build or partner with local project management capability from the outset.

SECTION 07

Conclusion

Nigeria's power sector deficit is, at its core, a capital deployment problem. The technology is available. The demand is proven beyond reasonable doubt. The regulatory environment has undergone its most significant reform in two decades. The returns available to structured private capital are among the highest of any infrastructure asset class globally.

What remains is the architecture of the deal: identifying the right investment model for the available capital, structuring agreements that appropriately allocate risk between private investors and public sector partners, and building the operational track record that unlocks institutional capital at scale.

This report is a first-principles framework for that architecture. It is intended to be a working document, not a static study, and is designed to support ongoing investor conversations, deal origination, and fund formation activity in Nigeria's energy infrastructure market.

For investment enquiries, deal structuring discussions, or fund formation conversations, please contact the author directly.

Disclaimer

This report is prepared for informational and discussion purposes only. It does not constitute investment advice, a solicitation to invest, or a guarantee of returns. All financial projections and return estimates are indicative and based on comparable market data. Actual results may differ materially from those projected. Prospective investors should conduct their own due diligence and consult qualified financial and legal advisors before making any investment decision.